



Department of Computer Science and Business Systems
Academic Year 2022 – 2023(Even Semester)

Degree, Semester & Branch: B.Tech., II & CSBS

Course Code & Title: AD3251 Data structures design

Name of the Faculty member: Ms.K.Usharani

Innovative Practice Description

- **Unit / Topic:** Unit 5 / Graph Traversal Techniques
- **Course Outcome:** CO 5 – Use efficient graph algorithms to solve graph problems
- **Topic Learning Outcome:** TLO 15 - Knowledge on graph traversals and discuss its applications.
- **Activity Chosen:** Think Pair Share
- **Justification:**
 1. Think Pair Share is an active learning and also a collaborative learning method which increase the students focus and involvement on learning the topic.
 2. Sharing their knowledge with their teams will improve their understanding in the topic and also their communication skill.
- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**
 1. **Material used :** Activity Sheet comprising a graph. Students has to find Breadth first traversal and Depth first traversal of the given graph. And also they have to draw corresponding BFS and DFS Tree.
 2. **Prerequisite :** Knowledge on the graph traversal techniques.
 3. Activity comprises of 3 parts – Think, Pair, Share
 4. **Think** – The students are asked to come up with the solution for the given problem individually.
 5. **Pair** – Students are combined as group of 2 members to discuss their knowledge and also to get clarity on the topic. They are asked to fill the worksheet after discussing with their team members.
 6. **Share** – Based on the solution given by the students, some of the teams were chosen to share their knowledge on the topic to their class students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
CO5	2	1	1	3	3	2	3

(1 – Low

2 – Moderate

3 – High)

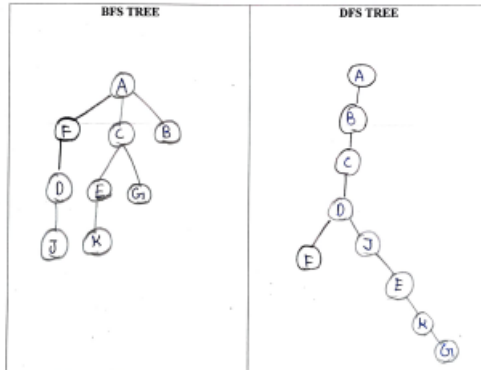
- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PO12	PSO3
	2	1	1	3	3	2	3
Justification for correlation	Apply the knowledge of engineering fundamentals like Stack and Queue concepts.	Identify and analyze complex engineering problems related to graph traversal.	Design solutions for complex engineering problems	Function effectively as an individual and as a member in a team	Communicate effectively on complex engineering activities with engineering community.	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning.	Apply appreciable knowledge of management skills cultured through real time practice.

- **Images / Screenshot of the practice:**



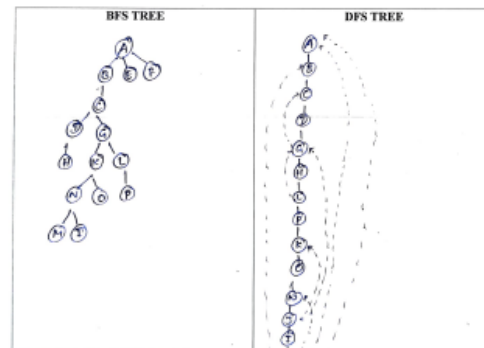
Q2. Traverse the graph in Breadth First order and Depth First order (Assume that a vertex's adjacencies are considered in alphabetical order). And also construct BFS Tree and DFS Tree.
Given Graph:



Name - Register Number :

1. Hoori Kishana . R [9536222 44017]
2. Uthaya Santhi . R V [9536222 44049]

Q1. Traverse the graph in Breadth First order and Depth First order (Assume that a vertex's adjacencies are considered in alphabetical order). And also construct BFS Tree and DFS Tree.
Given Graph:



Name - Register Number :

1. 9536222 44032 (S. Prityashwini)
2. 9536222 44022 (V. Teematchi)

• Reflective Critique:

❖ Feedback of practice from students and other stakeholders:

Students felt that they can able to understand and get clarity on the topic very easily by sharing their knowledge with their friends.

❖ Benefit of the practice:

Students were actively participated in this activity.

Students interaction with their peer members and the class during the activity was good and effective.

❖ Challenges faced in implementation:

Some students are hesitated to share their knowledge in the class. And they are motivated to come out of their fear.

References:

- ❖ <https://www.wgu.edu/heyteach/article/how-think-pair-share-activity-can-improve-your-classroom-discussions1704.html>
- ❖ <https://www.teachingexpertise.com/classroom-ideas/think-pair-share-activity/>
- ❖ <https://www.kent.edu/ctl/think-pair-share>